Cautare adversariala

noiembrie 2023

Outline

Strategii optimale
- Algoritmul MINIMAX
- $\alpha - \beta$ prunning

Decizii imperfecte, in timp real

Jocuri nedeterministe, cu sansa

Jocuri cu observabilitate partiala

Jocuri

Game-theory - ramura a economiei: medii multiagent, impactul unui agent asupra celuilalt este semnificativ
Medii multi-agent:

- fiecare agent trebuie sa considere actiunile celorlalti agenti
- oponent imprevizibil, incontrolabil
- mediu competitiv

$\Rightarrow$ cautare adversariala
Solutia este o **strategie (politica)** care specifica mutarea pentru fiecare mutare posibila a oponentului
Obs:In probleme de cautare, solutia este o secventa de actiuni care garanteaza obtinerea scopului

Tipuri de jocuri

	deterministic	chance
perfect information	chess, checkers, go, othello	backgammon monopoly
imperfect information	battleships, blind tictactoe	bridge, poker, scrabble nuclear war

- determinist sau stocastic?
- un jucator, doi sau mai multi?
- zero-sum game? - payoff-ul total al tuturor jucatorilor e acelasi pt fiecare instanta de joc
- informatii perfecte (se cunoaste intreaga stare)

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Starea jocului usor de reprezentat, efectele actiunilor sunt definite de reguli clare; valabil si pt hockey?

Scurt istoric

- 1950 - Turing - primul program de sah
- dame si Othello - masina mai buna decat omul
- sah si table - masina bate uneori campionii umani
- Go - un campion uman a fost invins

- sah: branching factor ≈ 35, considerand 50 de mutari $\rightarrow$ 35^{100} noduri in arborele de cautare
- e necesara abilitatea de a lua decizii chiar daca decizia optima nu se poate calcula

Zero sum games



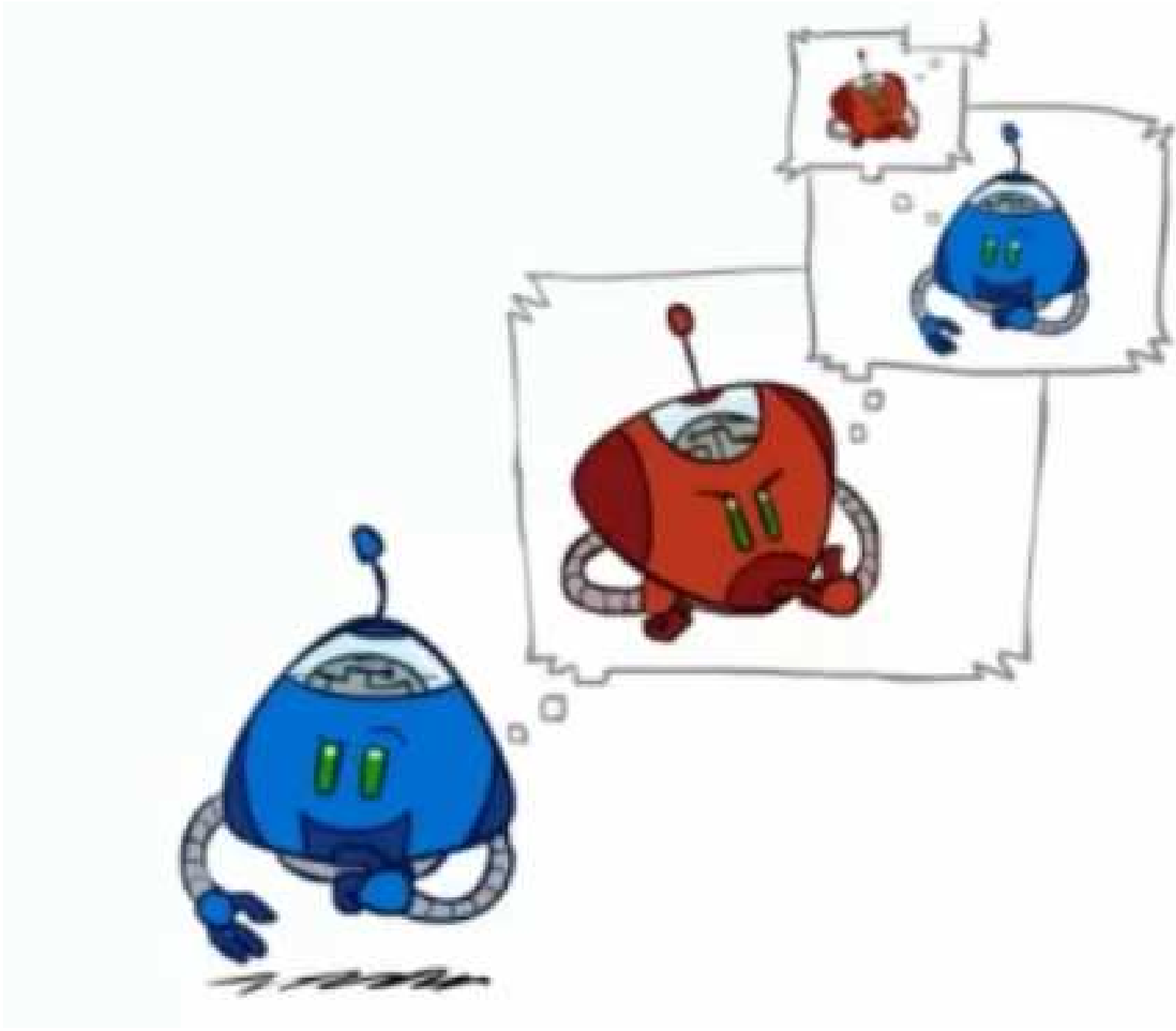
- Jocuri cu suma zero
  - agentii au utilitati (evaluari ale rezultatelor) opuse (sau)
  - suma utilitatilor este constanta
  - un agent doreste maximizarea utilitatii, celalalt minimizarea
  - mediu adversarial pur competitiv
- Jocuri generale
  - agentii au utilitati independente
  - cooperare, indiferenta, competitie

Formularea problemei

Jocuri cu doi jucatori; la finalul jocului punctele sunt ale castigatorului, penalitatile sunt ale pierzatorului

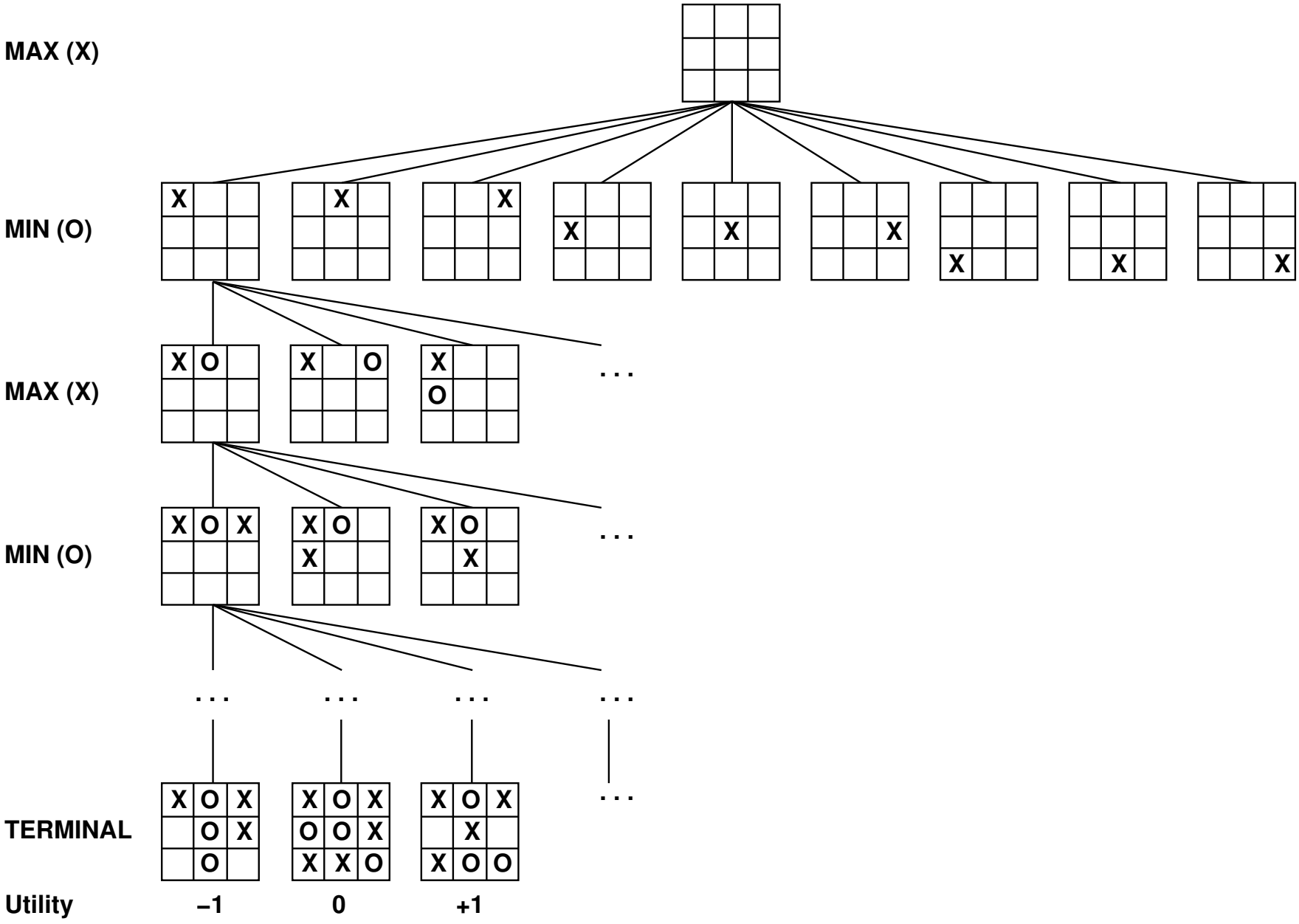
- stare initiala: s_0
- jucatorul care trebuie sa mute intr-o stare: PLAYER(s)
- actiuni legale intr-o stare: ACTIONS(s)
- modelul de tranzitie: RESULT(s,a)
- terminal test: TERMINAL-TEST(s)
- o functie de utilitate - asociaza o valoare numerica starilor terminale: UTILITY(s,p)

Sah: functia de utilitate: 1/2 - remiza, 1 castig, 0 pierdere



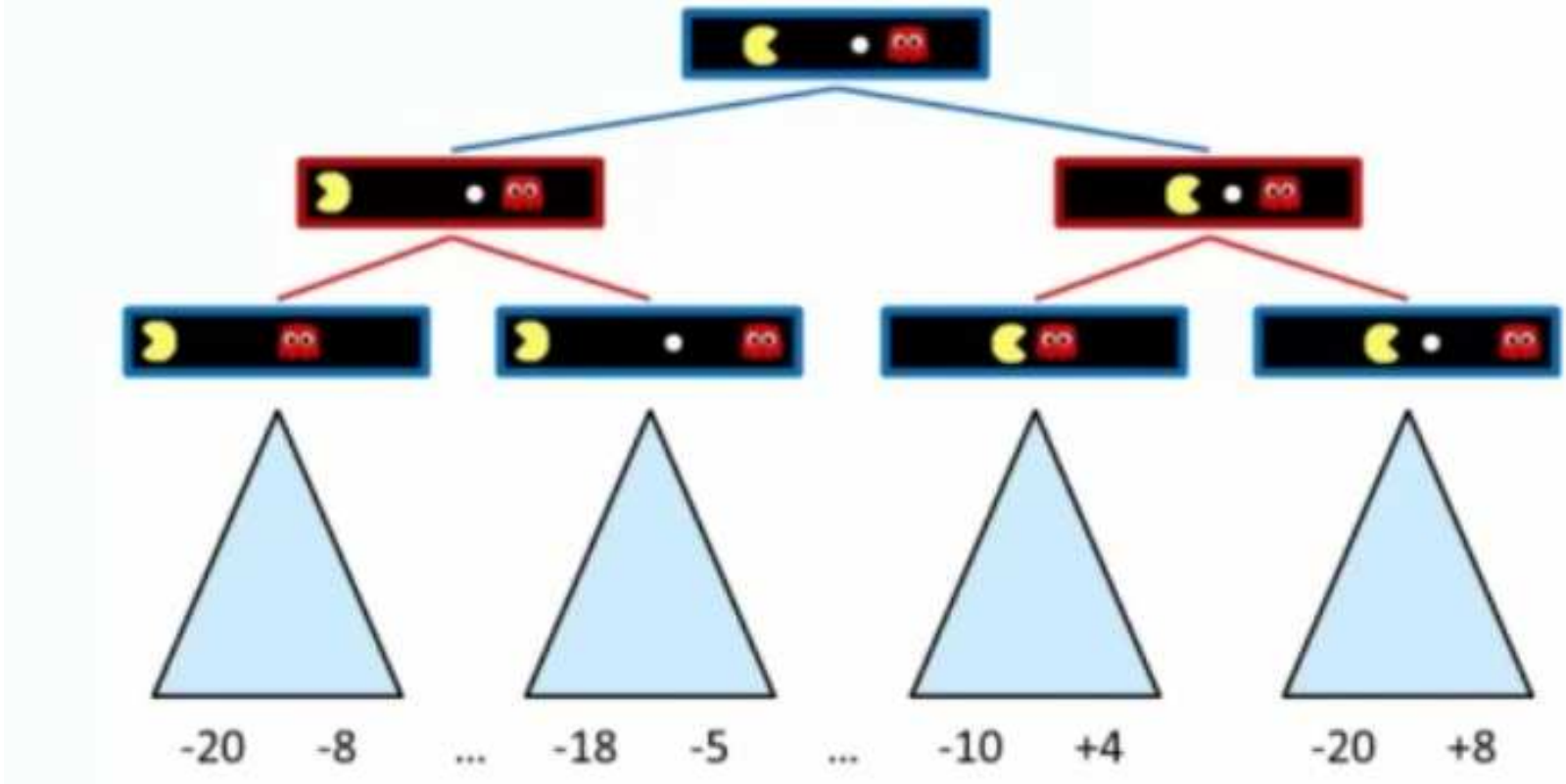
imagine preluata de pe edX

Arbore de joc pentru X-0



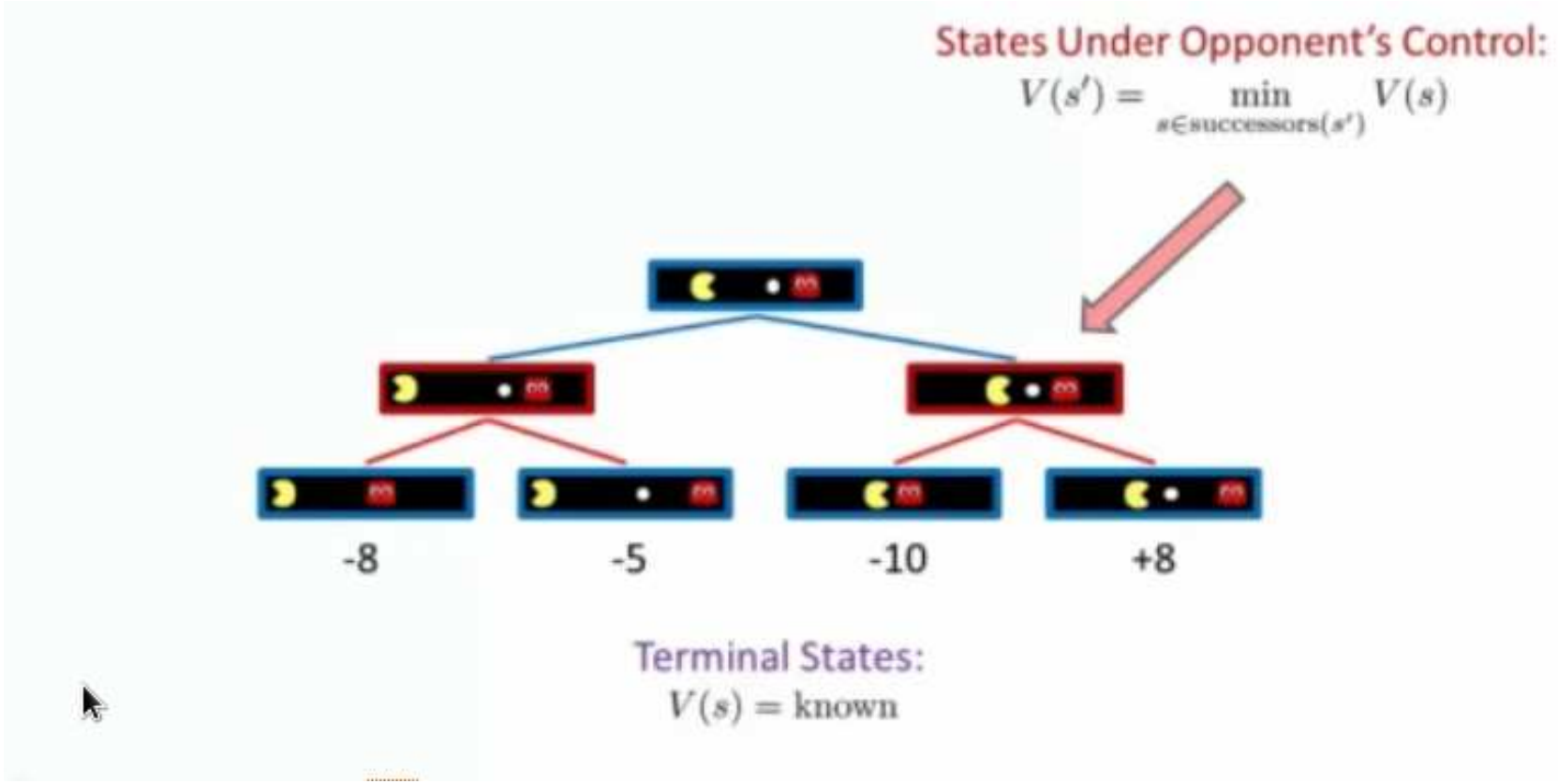
Arbore de joc in medii cu doi agenti, deterministic, mutari pe rand

Agentii sunt in acord cu ce se poate face, dar in dezacord cu ce vor alege sa faca
possible futures + who controls the actual node?



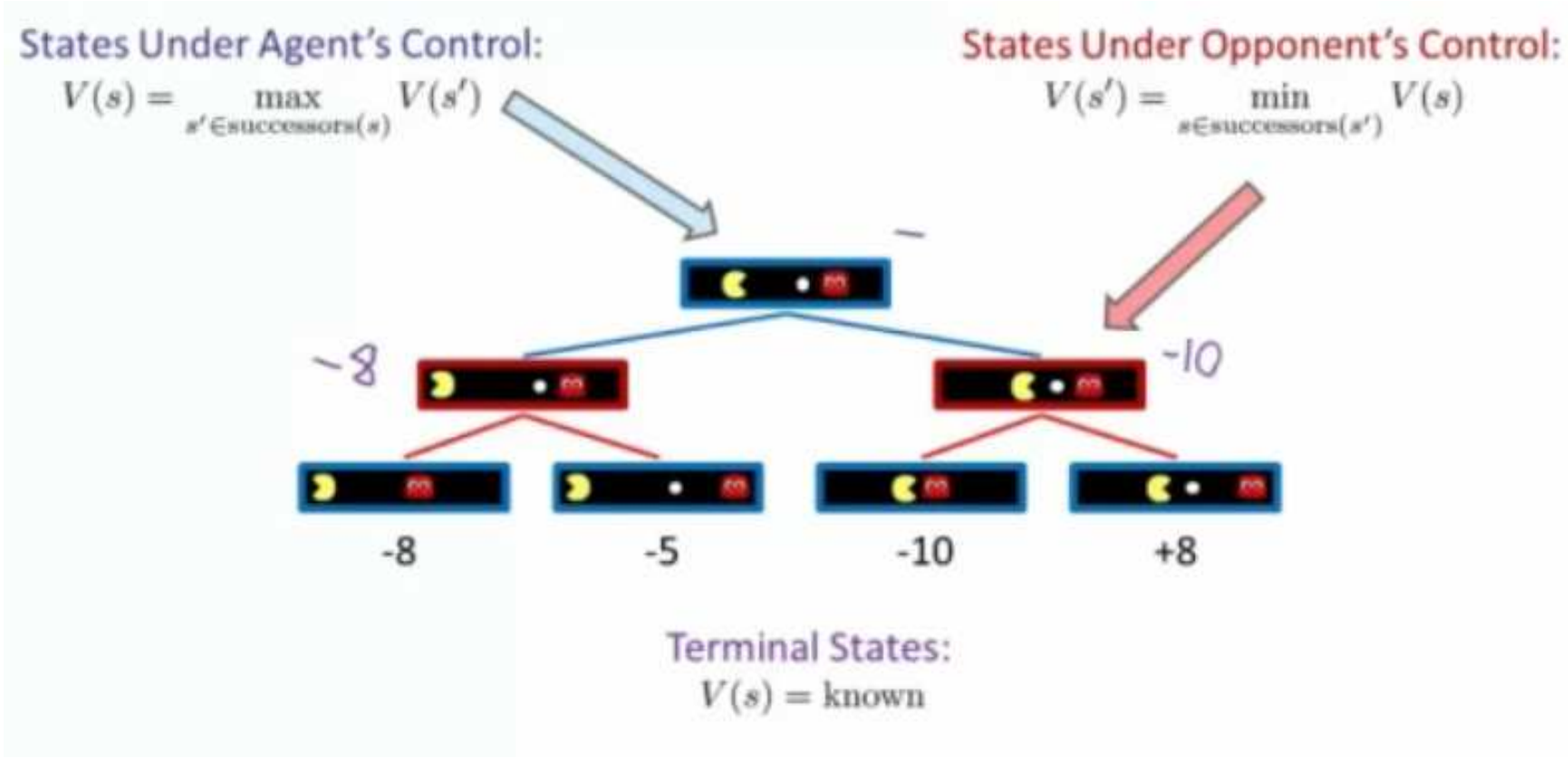
imagine preluata de pe edX

Arbore de joc in medii cu doi agenti, deterministic, mutari pe rand



imagine preluata de pe edX

Arbore de joc in medii cu doi agenti, deterministic, mutari pe rand



Ce se poate face cel mai bine, dar luand in considerare actiuni optime ale adversarului $\rightarrow$ **strategie optimala**

Min, Max, Valoare MiniMax a starii

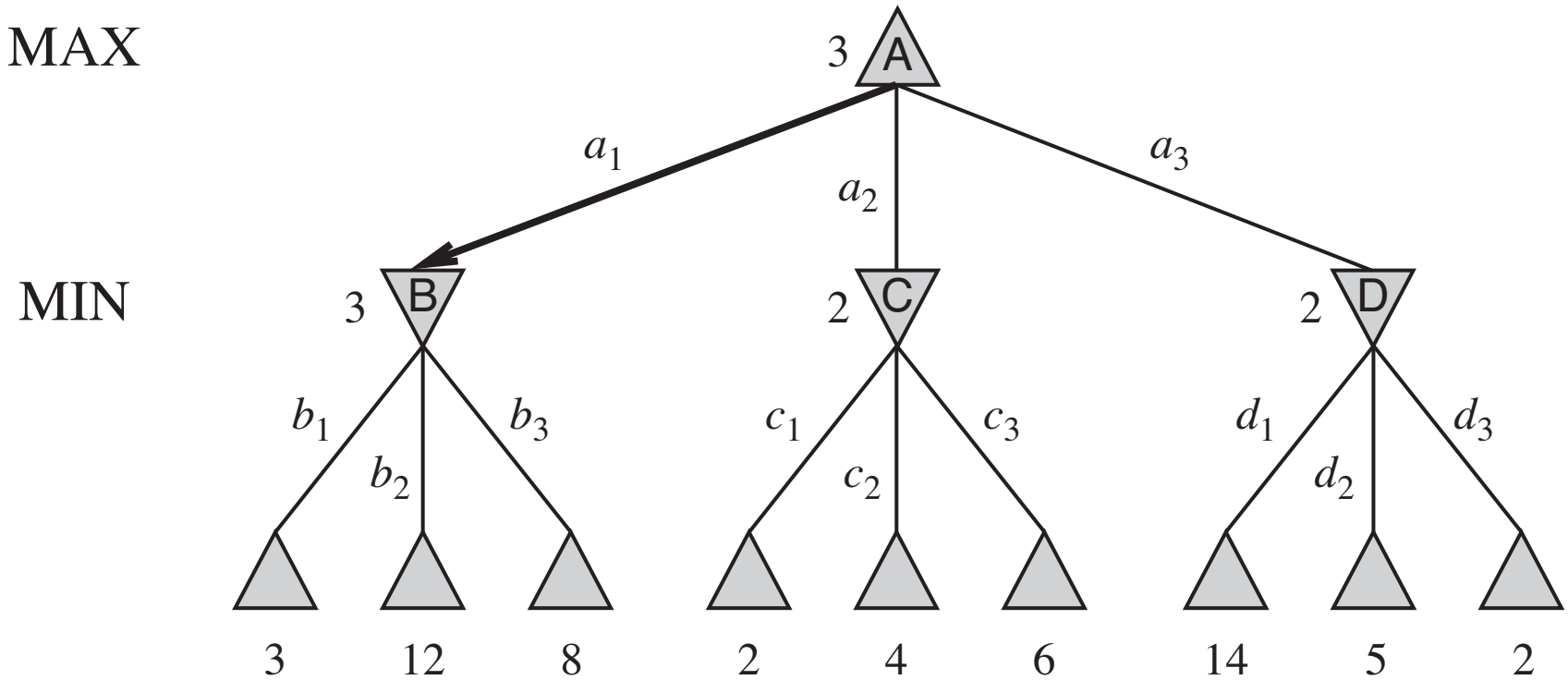
imagine preluata de pe edX

Algoritmul MINIMAX

- Strategie de joc perfect pentru jocuri deterministe, cu informatie perfecta
- Alege mutarea catre pozitia cu cea mai mare valoare minimax = cel mai bun rezultat ce se poate obtine impotriva unui jucator perfect

MINIMAX(s)=
=UTILITY(s) daca TERMINAL-TEST(s)
$=max_{a \in Actions(s)}$ MINIMAX(RESULT(s,a)) daca PLAYER(s)= MAX
$=min_{a \in Actions(s)}$ MINIMAX(RESULT(s,a)) daca PLAYER(s)=MIN

Exemplu



Algoritmul MINIMAX

```
function MINIMAX-DECISION (state) returns an action
   return argmax_{a∈ACTIONS(s)} MIN-VALUE(RESULT(state, a))
function MAX-VALUE(state) returns a utility value
 if TERMINAL-TEST(state) then return UTILITY (state)
 v ← −∞
 for each a in ACTIONS(state) do
   v ← MAX(v, MIN-VALUE(RESULT(s, a)))
return v

function MIN-VALUE(state) returns a utility value
 if TERMINAL-TEST(state) then return UTILITY (state)
 v ← ∞
 for each a in ACTIONS(state) do
   v ← MIN(v,MAX-VALUE(RESULT(s, a)))
return v
```

Proprietati ale minimax

Complet? Da, daca arborele este finit

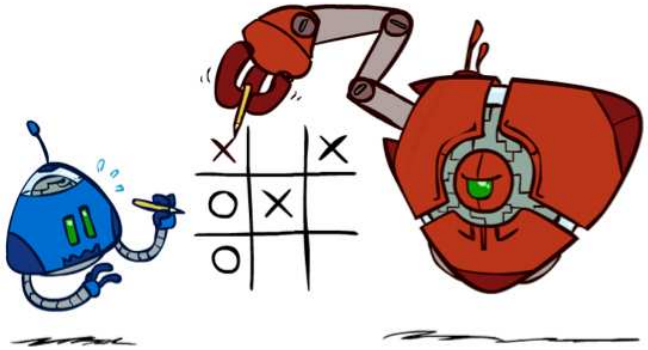
Optimal? Da, impotriva unui oponent optimal.

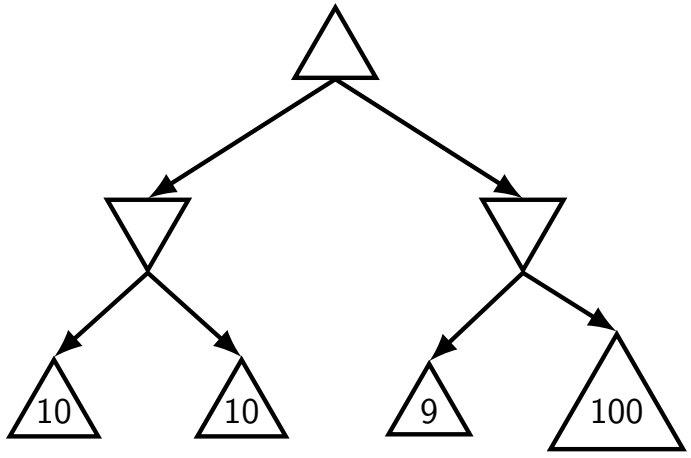
Complexitatea timpului $O(b^m)$

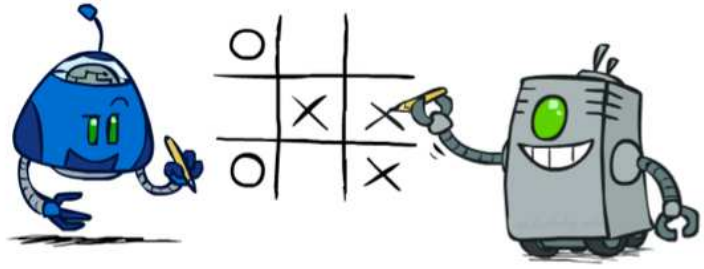
Complexitatea spatiului $O(bm)$ (explorare depth-first)

pt sah: $b \approx 35, m \approx 100 \rightarrow$ solutie exacta nefezabila

continuare Proprietati ale MINIMAX

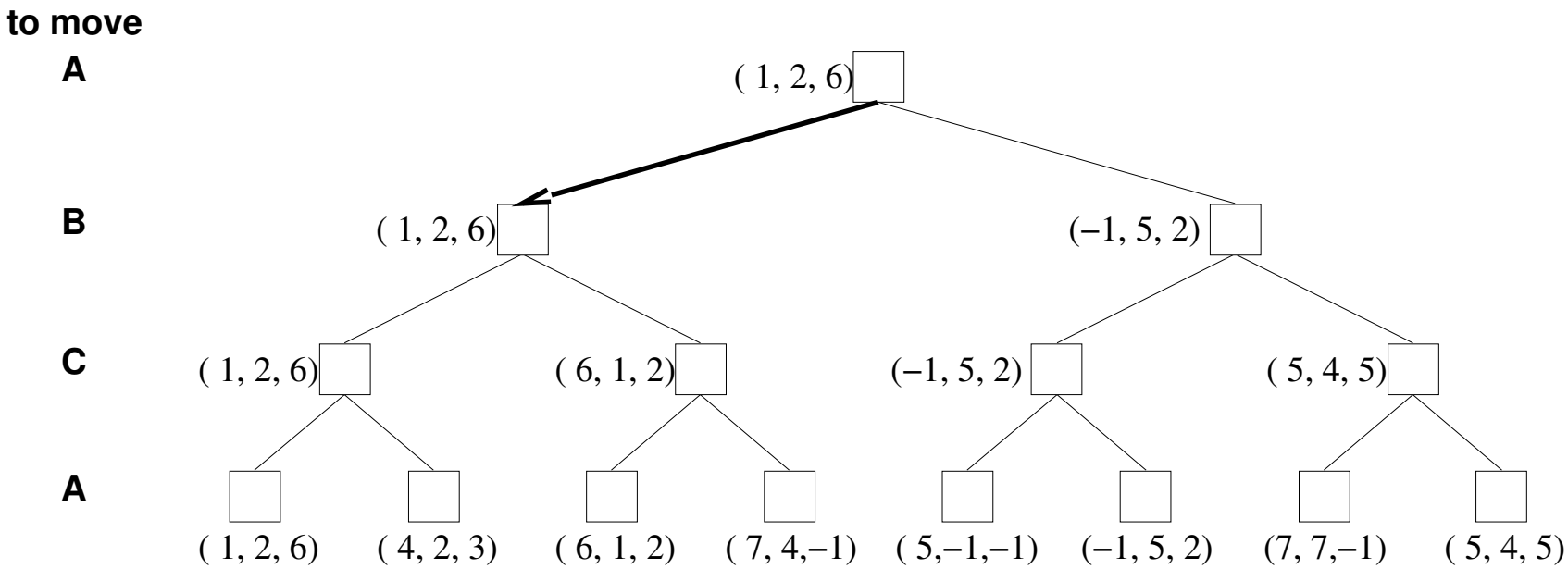






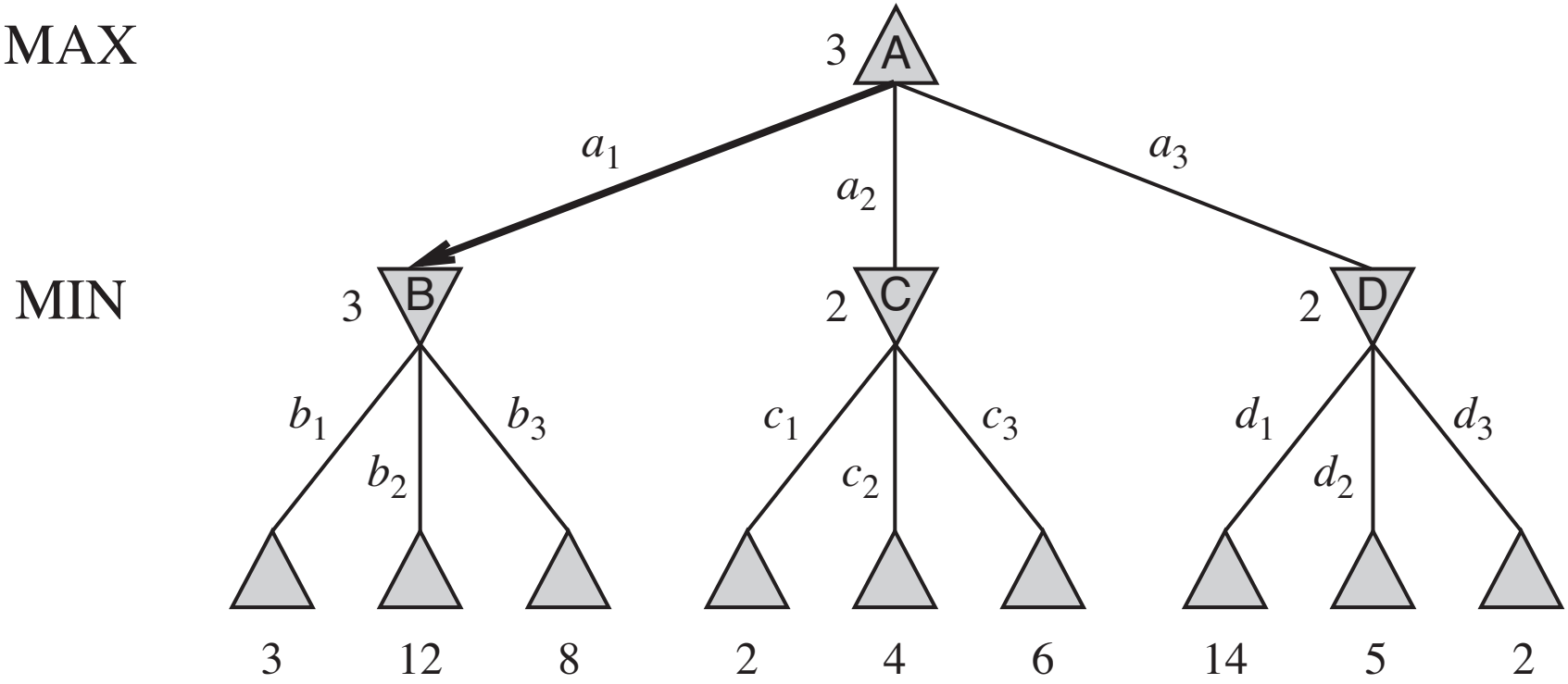
see pacman minimax trapped

Mai mult de 2 agenti



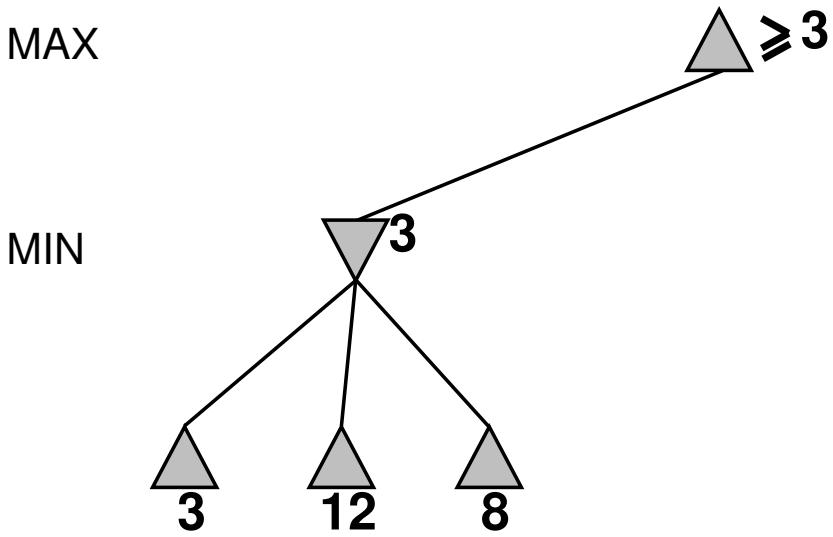
exemple pacman

Exemplu

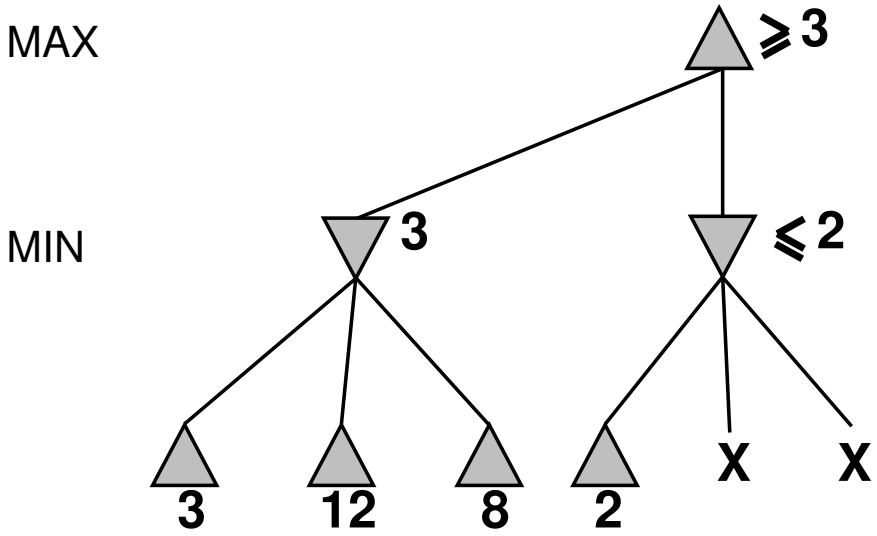


Trebuie explorate toate nodurile???

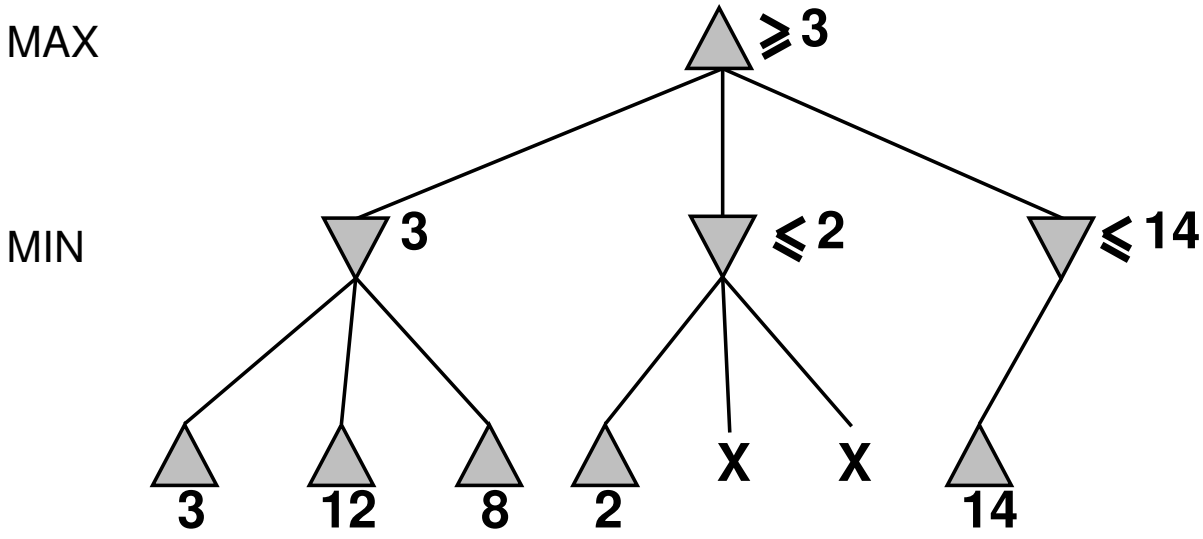
Exemplu $\alpha - \beta$ prunning



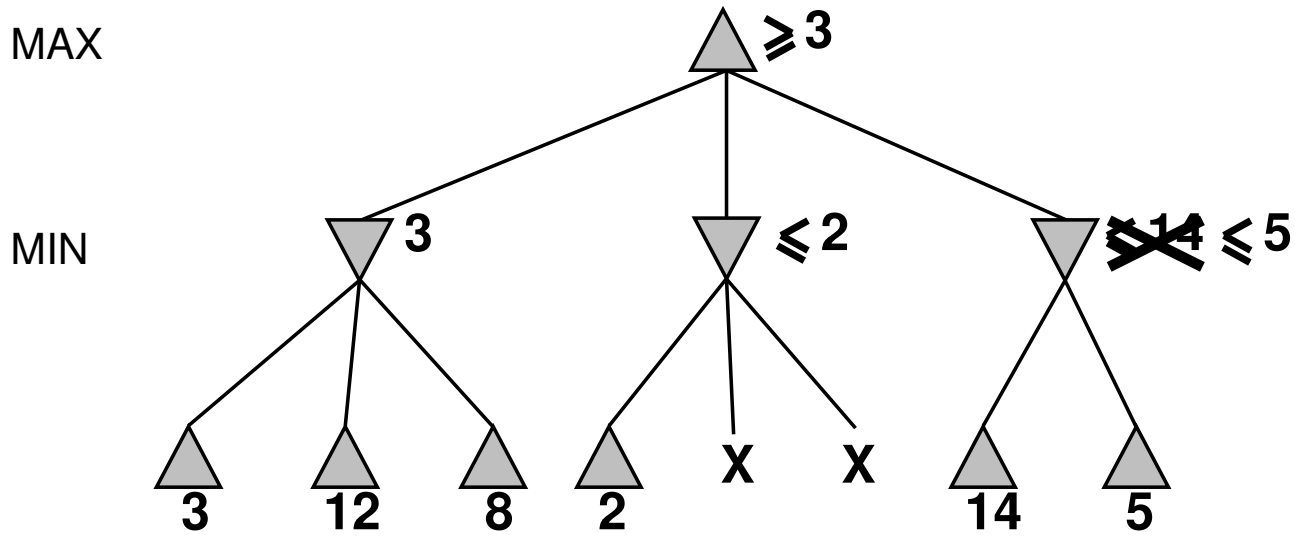
Exemplu $\alpha - \beta$ prunning



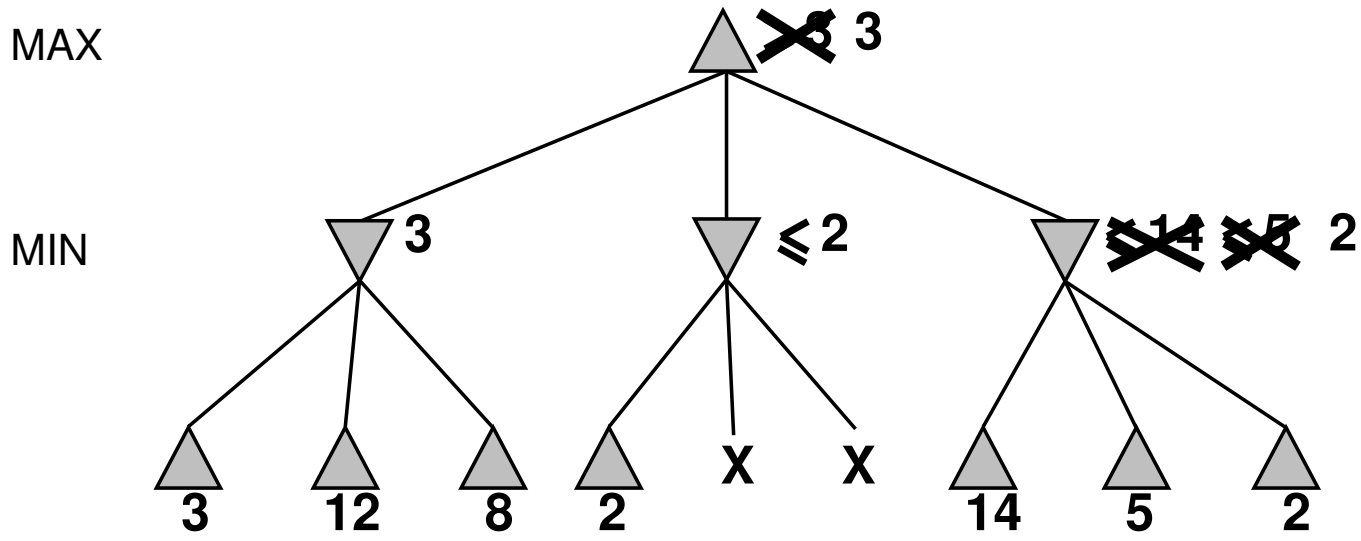
Exemplu $\alpha - \beta$ prunning



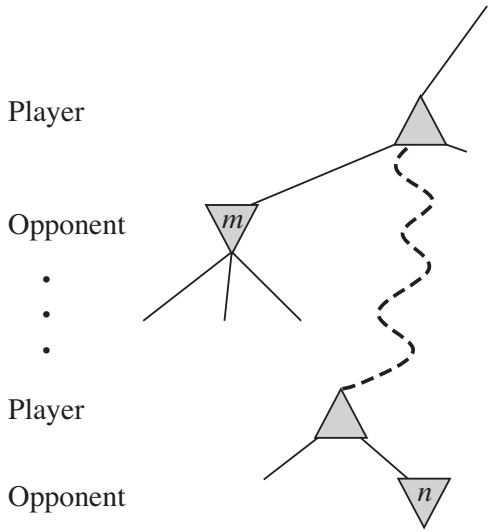
Exemplu $\alpha - \beta$ prunning



Exemplu $\alpha - \beta$ prunning



De ce $\alpha - \beta$



Daca *m* este mai bun decat n pt Player, Player va evita *n*, deci copiii lui *n* nu mai trebuie considerati

- α este cea mai buna valoare (cea mai mare) gasita pana acum la orice punct de decizie de-a lungul caii pt MAX
- β cea mai buna valoare (cea mai mica) gasita pana acum la orice punct de decizie de-a lungul caii pentru pt MIN

Algoritmul $\alpha - \beta$

```
function ALPHA-BETA-SEARCH (state) returns an action
  v ← MAX-VALUE(state, −∞, ∞)
return the action in ACTIONS(state ) with value v

function MAX-VALUE(state , α, β) returns a utility value
if TERMINAL-TEST(state) then return UTILITY (state)
v ← −∞
for each a in ACTIONS(state) do
 v ← MAX(v , MIN-VALUE(RESULT(s,a), α, β))
 if v ≥ β then return v
 α ← MAX(α, v )
return v

function MIN-VALUE(state , α, β) returns a utility value
if TERMINAL-TEST(state) then return UTILITY (state)
v ← +∞
for each a in ACTIONS(state) do
 v ← MIN(v, MAX-VALUE(RESULT(s,a),α, beta))
 if v ≤ α then return v
 β ← MIN(β, v )
```

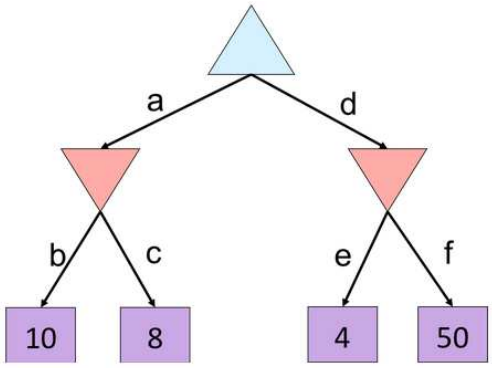
Reluare $\alpha - \beta$ prunning

- pt MAX: v e max; if $v \geq \beta$ prunning; $\alpha \leftarrow max(\alpha, v)$
- pt MIN: v e min; if$v \leq \alpha$ prunning; $\beta \leftarrow min(\beta, v)$

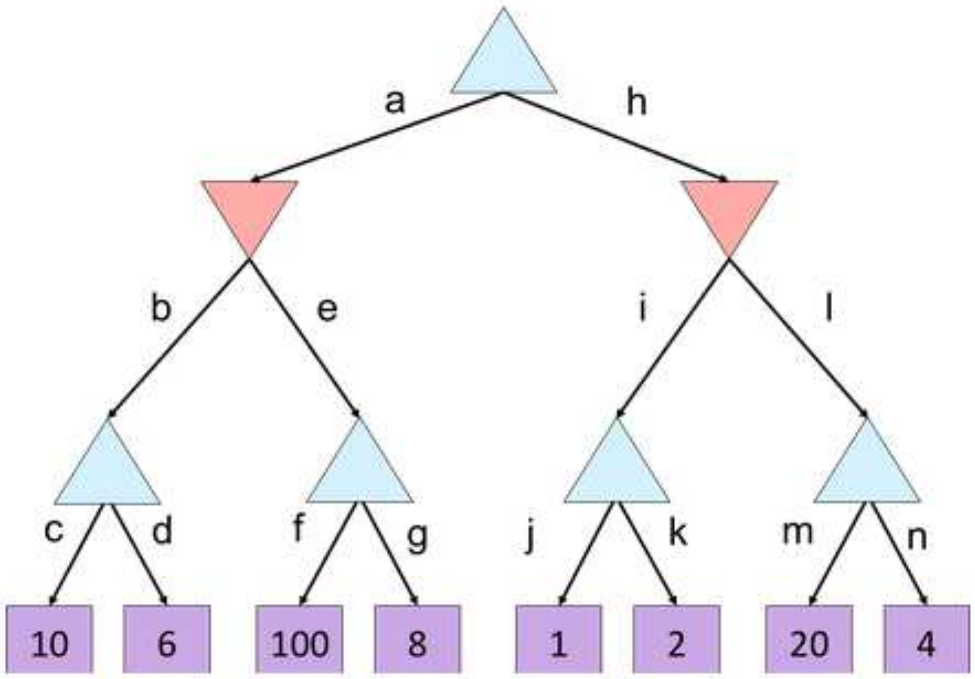
Exemplu $\alpha - \beta$ prunning

http://homepage.ufp.pt/jtorres/ensino/ia/alfabeta.html

quiz

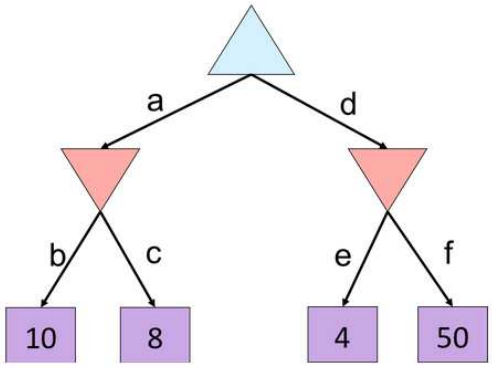
1. 

a, b, c, d, e, f, nici o ramura

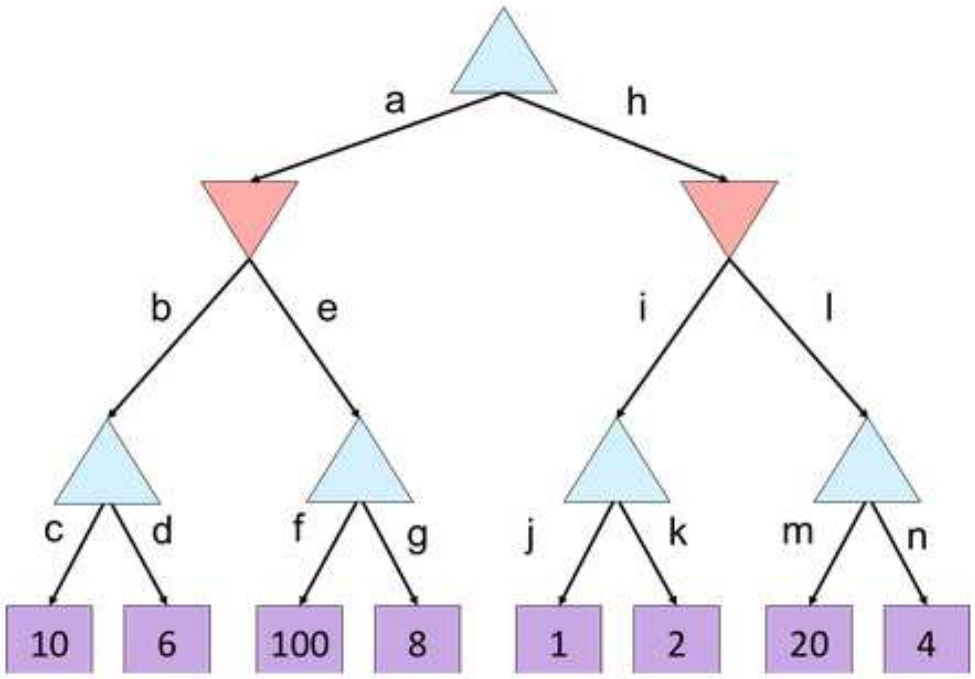
2. 

a, b, c, d, e, f, g, h, i, j, k,l, m, n, nici o ramura

quiz

1. 

a, b, c, d, e, f, nici o ramura

2. 

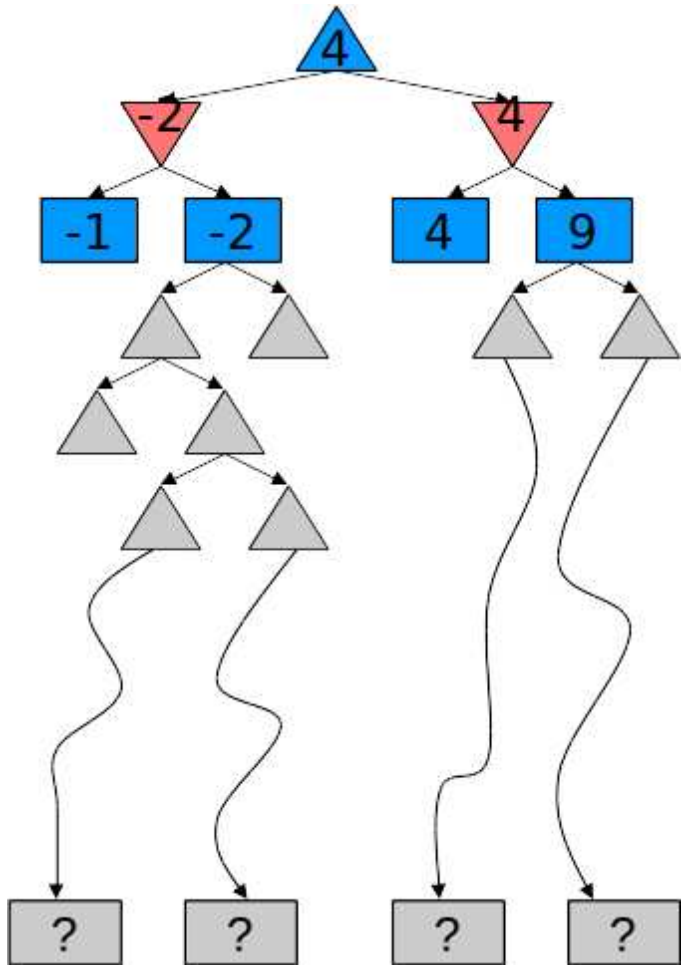
a, b, c, d, e, f, g, h, i, j, k,l, m, n, nici o ramura

Proprietatile algoritmului $\alpha - \beta$

- Taierea ramurilor nu afecteaza rezultatul final
- o ordonare buna a mutarilor imbunatateste eficienta taierii
- cu o ordonare buna, complexitatea timpului $= O(b^{m/2})$
- din pacate 35^{50} este tot prea mare

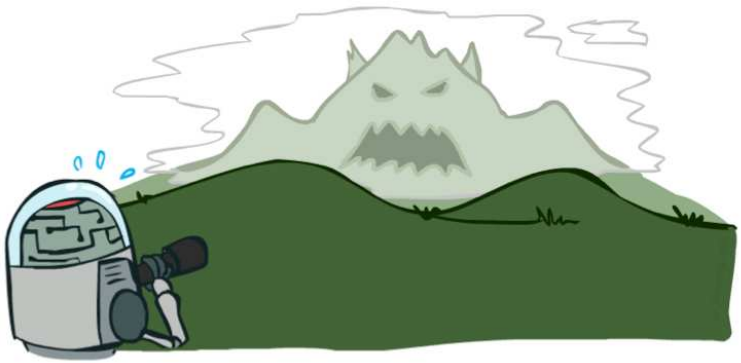
Decizii imperfecte, in timp real: Limitele resurselor

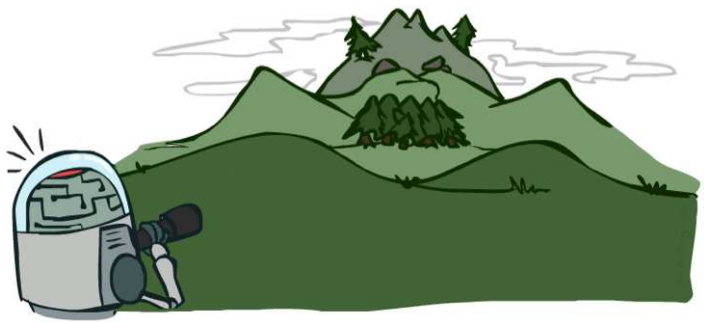
- utilizarea CUTOFF-TEST in locul lui TERMINAL-TEST
  - limitarea adancimii - depth limited search, iterative deepening search
- utilizarea EVAL in loc de UTILITY
  - functia de evaluare **estimeaza** dezirabilitatea unei pozitii - utilitatea asteptata a unui joc dintr-o anumita pozitie
  - functie de evaluare e imperfecta



Functia de evaluare

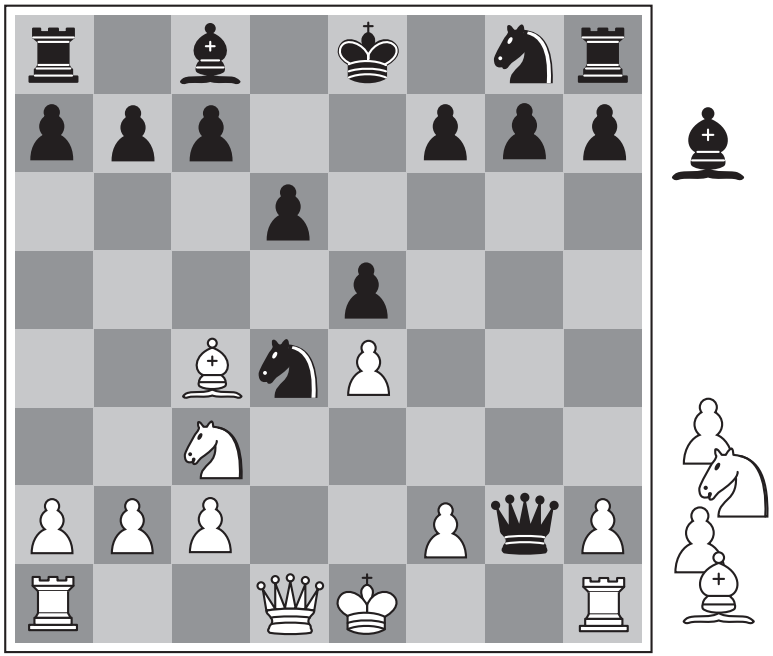
- intotdeauna sunt imperfecte, DAR
- trebuie sa ordoneze starile terminale in acelasi fel ca functia utilitate
- calcularea ei trebuie sa nu dureze mult timp
- pentru starile neterminale trebuie sa fie corelata cu sansele reale de castig



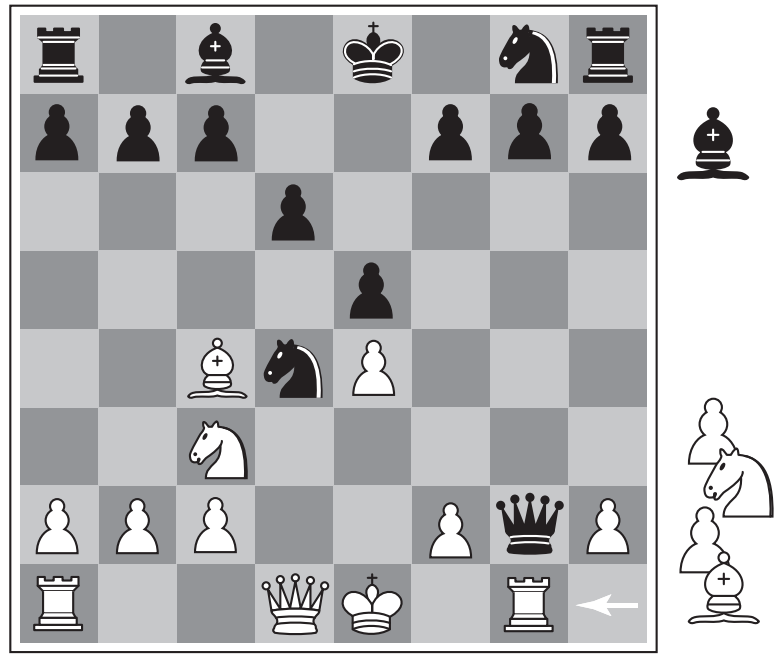


see pacman - horizon

Functia de evaluare bazata pe caracteristici



(a) White to move



(b) White to move

Suma ponderata liniara (weighted linear function) a unor caracteristici $Eval(s) = w_1 f_1(s) + w_2 f_2(s) + ... + w_n f_n(s)$

Caracteristici:

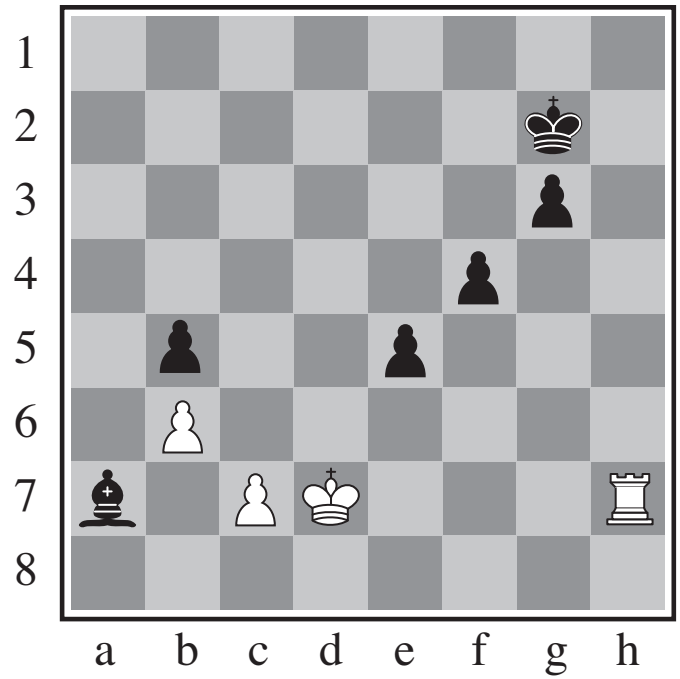
- piesele disponibile cu valoarea lor materiala: un pion = 1, un cal sau nebun=3, un turn=5, regina=9
- o structura buna a pionilor, siguranta regelui = jumatate de pion

Evaluare bazata pe caracteristici

- weighted linear function
  - Suma lineara ponderata e usor de calculat, dar presupune independenta valorilor caracteristicilor: nebunii sunt mai valorosi in finaluri, pt ca au mai mult spatiu de manevra; doi nebuni valoreaza mai mult decat de doua ori valoarea unui nebun
  - Caracteristicile si ponderile nu tin de regulile jocului, ci de experienta.
- expected value: categorii sau clase echivalente de stari: ex: toate starile cu doi pioni vs un pion
  proportie a starilor cu rezultatul lor:
  - 72% conduc la castig
  - 20% conduc la pierdere
  - 8% conduc la remiza

$$0.72 * 1 + 0.2 * (-1) + 0.08 * 0.5$$

Probleme ale Trunchierii cautarii



- quiescent search: trunchierea se face doar pt pozitii care nu produc schimbari extreme in valoare in viitorul apropiat
  ex: nonquiescent position: pozitii din care se pot face capturi favorabile
- efectul de orizont: se amana mutari ale adversarului care produc stricaciuni serioase
  ex: nebunul negru este pierdut - turnul alb; negrul propune o secventa de actiuni care impinge capturarea dupa orizont: sah cu pionul de la e5

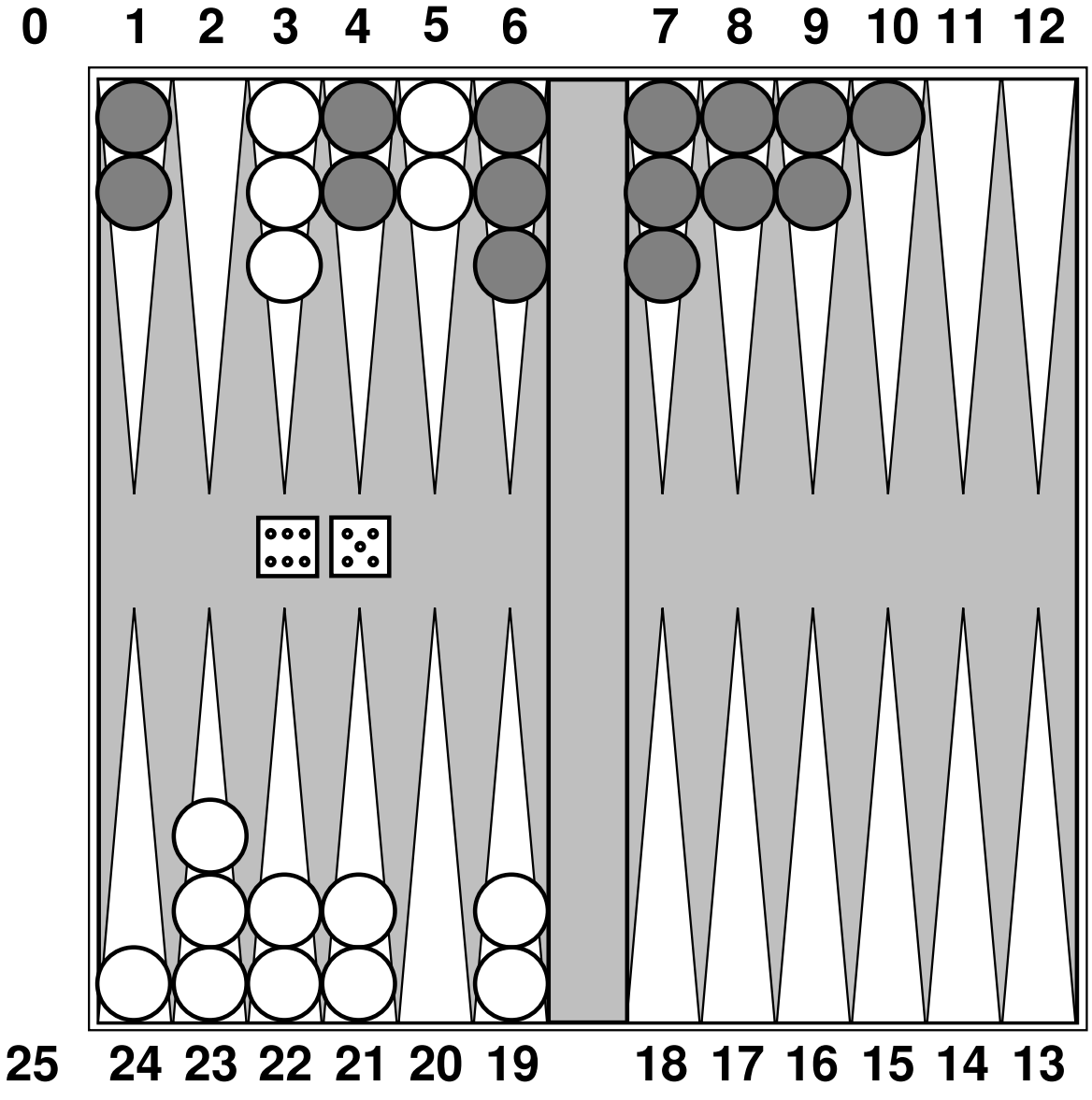
Jocuri deterministe in practica

- Checkers: Chinook ended 40-year-reign of human world champion Marion Tinsley in 1994. Used an **endgame database** defining perfect play for all positions involving 8 or fewer pieces on the board, a total of 443,748,401,247 positions.
- Chess: Deep Blue defeated human world champion Gary Kasparov in a six-game match in 1997. Deep Blue searches **200 million positions** per second, uses very sophisticated evaluation, and undisclosed methods for **extending some lines of search up to 40 ply**.
- Othello: human champions refuse to compete against computers, who **are too good**.
- Go: $b > 300$, so most programs use pattern knowledge bases to suggest plausible moves.
  `https://deepmind.com/research/case-studies/alphago-the`

MonteCarlo Tree

AIMA - utilizat in Go

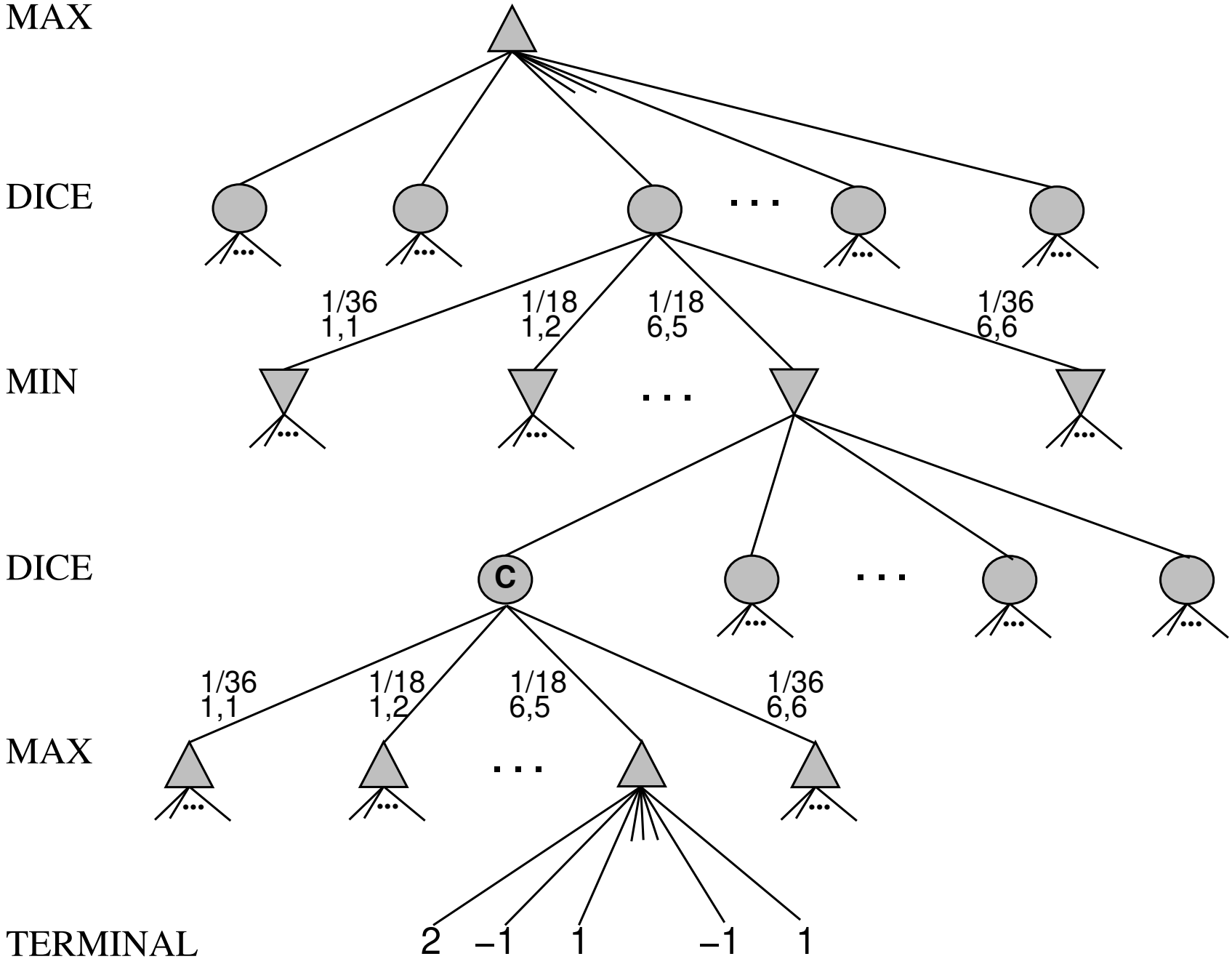
Jocuri cu sansa



legal moves:

(5-10,5-11), (5-11,19-24), (5-10,10-16), (5-11,11-16)

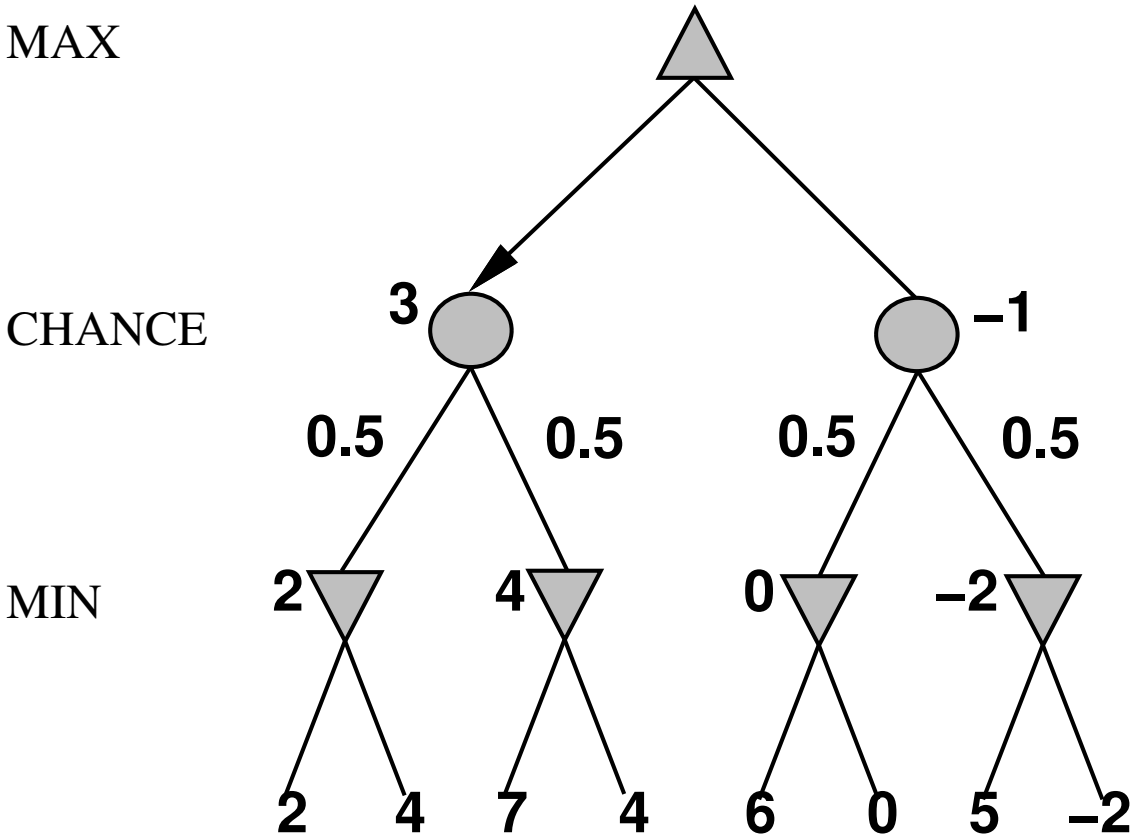
Arbore pentru jocuri cu sansa



Jocuri nedeterministe

Sansa: introdusa de zaruri, de amestecarea cartilor...: table
Incertitudinea nu e determinata doar de un adversar, ci si de sansa

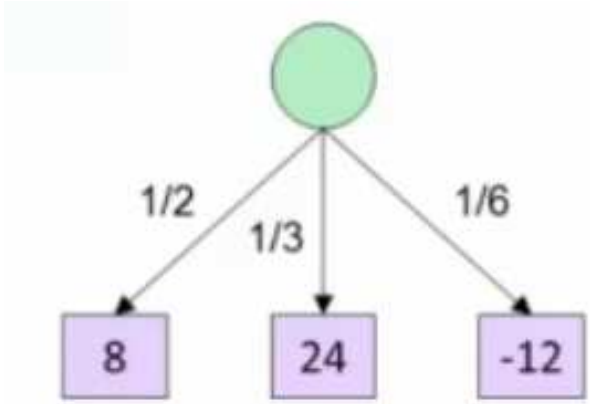
- In arborele jocului se introduc noduri sansa



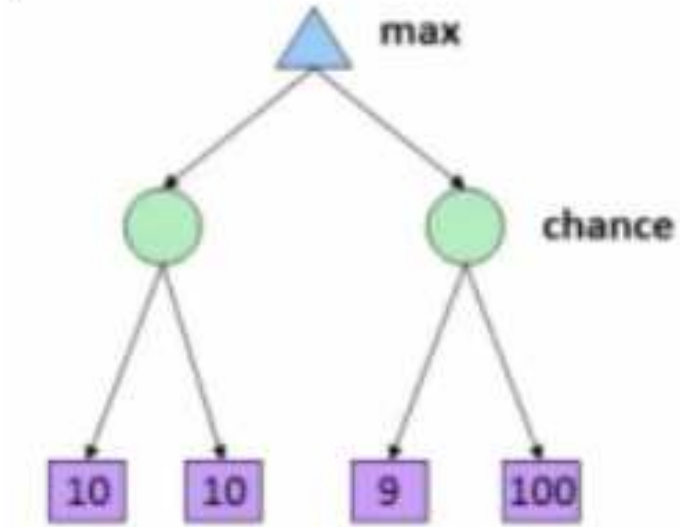
Expectiminimax

EXPECTIMINIMAX = MINIMAX + noduri sansa
EXPECTIMINIMAX(n)=
=UTILITY(n) daca TERMINAL-TEST(s)
$=max_a$ EXPECTIMINIMAX(RESULT(s,a)) daca PLAYER(s)= MAX
$=min_a$ EXPECTIMINIMAX(RESULT(s,a)) daca PLAYER(s)=MIN
$=\Sigma_r$ P(r)EXPECTIMINIMAX(RESULT(s,r))daca PLAYER(s)=CHANCE

Valoare asteptata

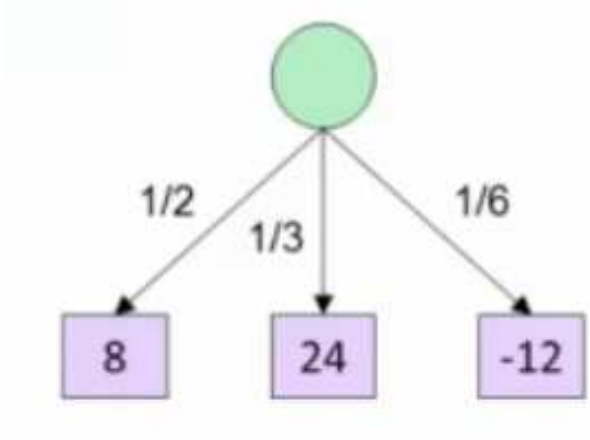


v=1/2 * 8+ 1/3 * 24 + 1/6 * -12

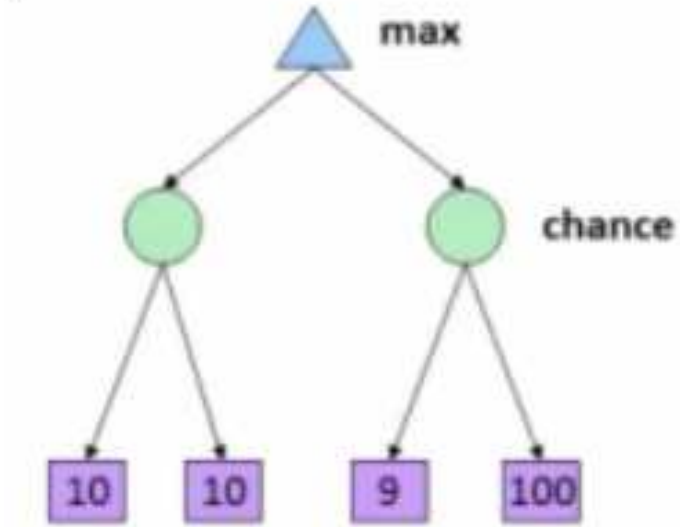


Care e actiunea buna? stanga sau dreapta?

Valoare asteptata



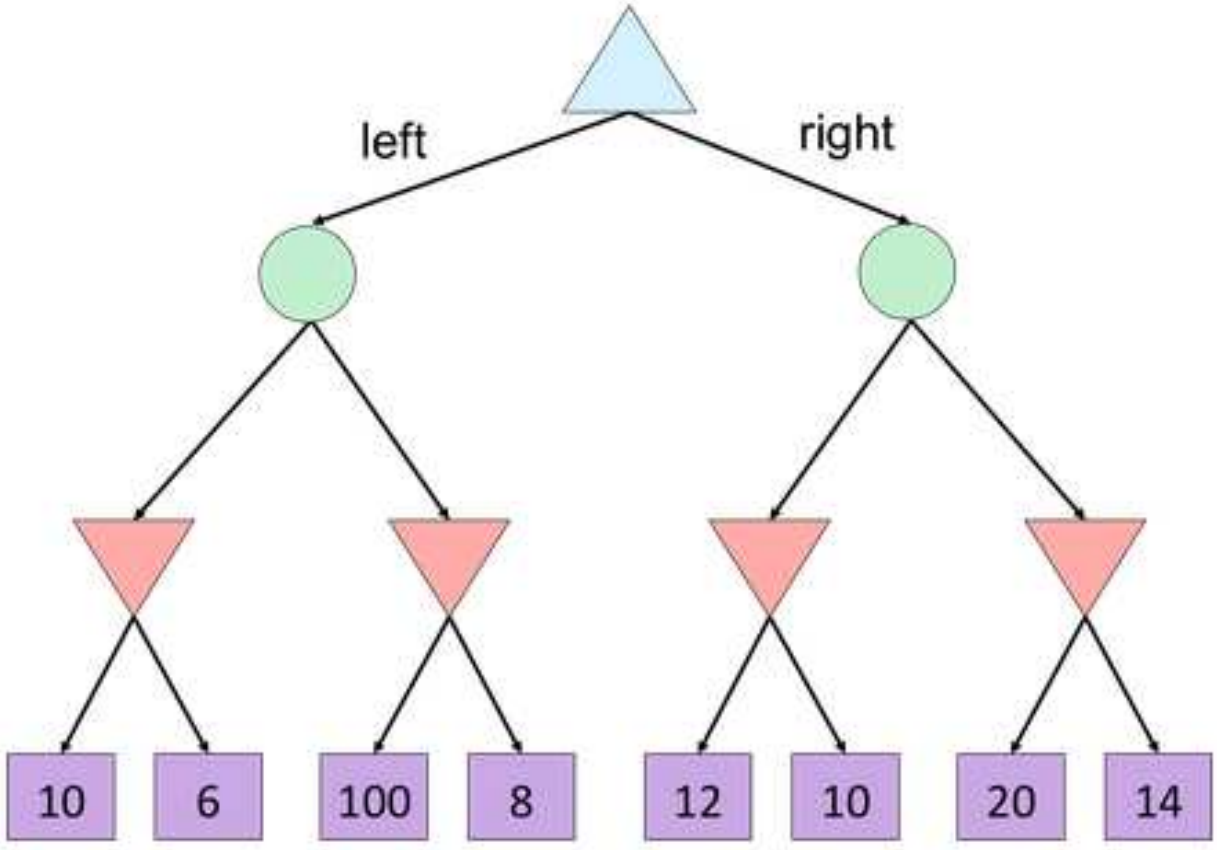
v=1/2 * 8+ 1/3 * 24 + 1/6 * -12



daca toate ramurile nodurilor sansa au aceeasi probabilitate mutarea right e mai buna

Quiz

Se aplica expectiminimax pe arborele de joc. Ramurile nodurilor au aceeasi probabilitate

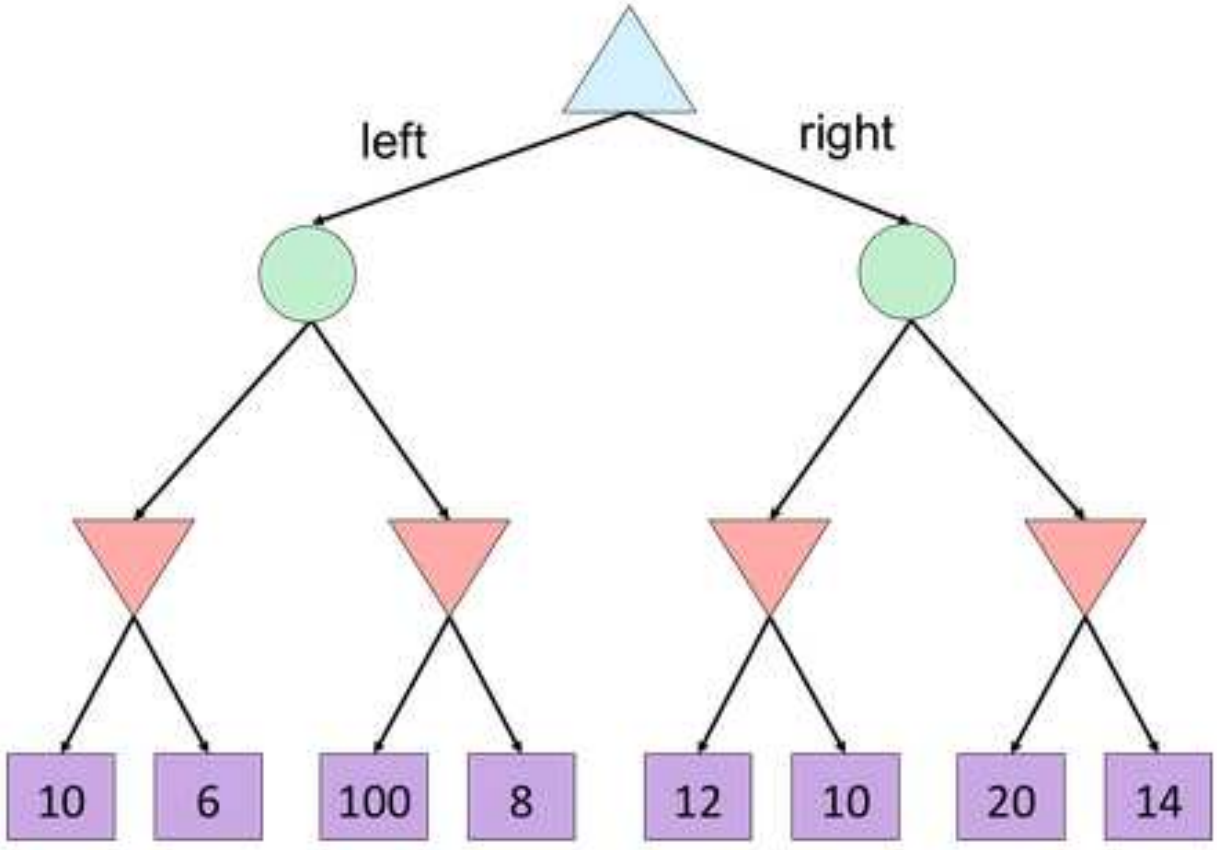


Care e valoarea jocului?

Care actiune e optimala pt max?

Quiz

Se aplica expectiminimax pe arborele de joc. Ramurile nodurilor au aceeasi probabilitate



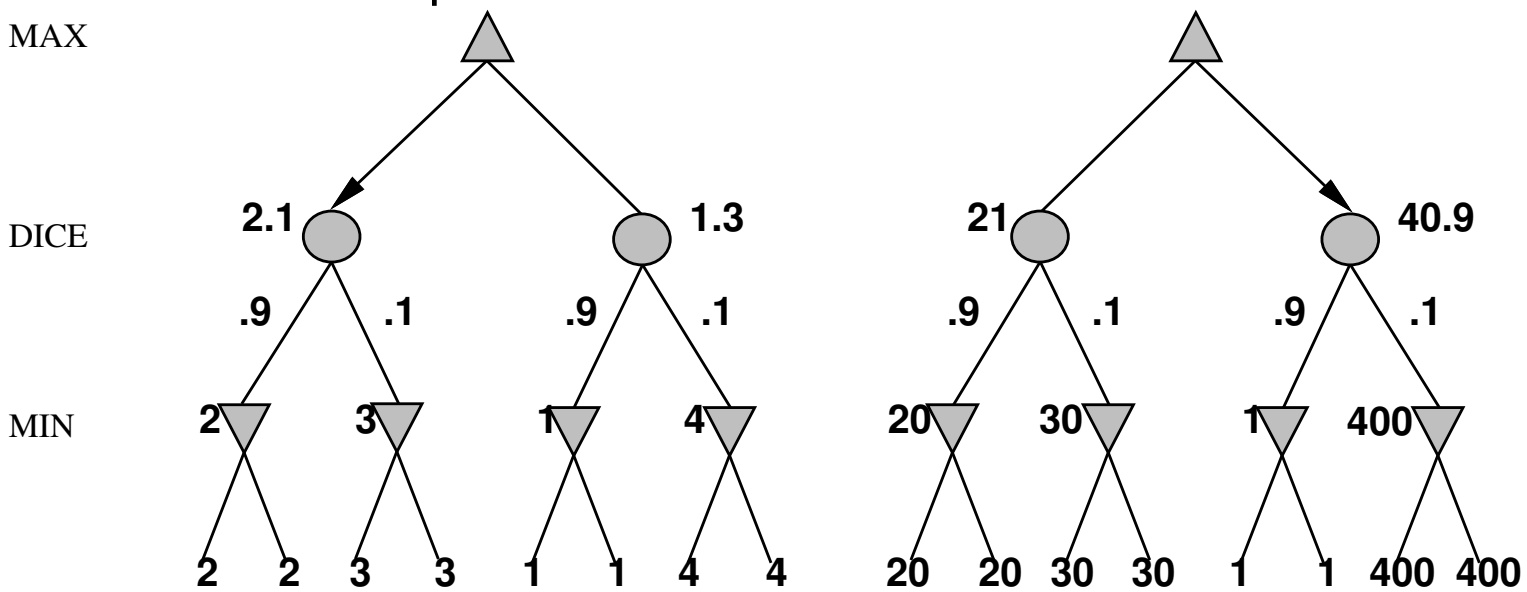
Care e valoarea jocului? **12**

Care actiune e optimala pt max? **right**

Valorile functiei de evaluare

In jocurile deterministe: functia de evaluare trebuie doar sa dea valori mai mari pozitiilor mai bune

In jocurile cu sansa: valorile conteaza, trebuie sa fie proportionale cu utilitatea asteptata



Jocuri cu observabilitate partiala

possible states; anunturi: Check by direction, Capture on square X
mutare pt fiecare secventa de perceptii

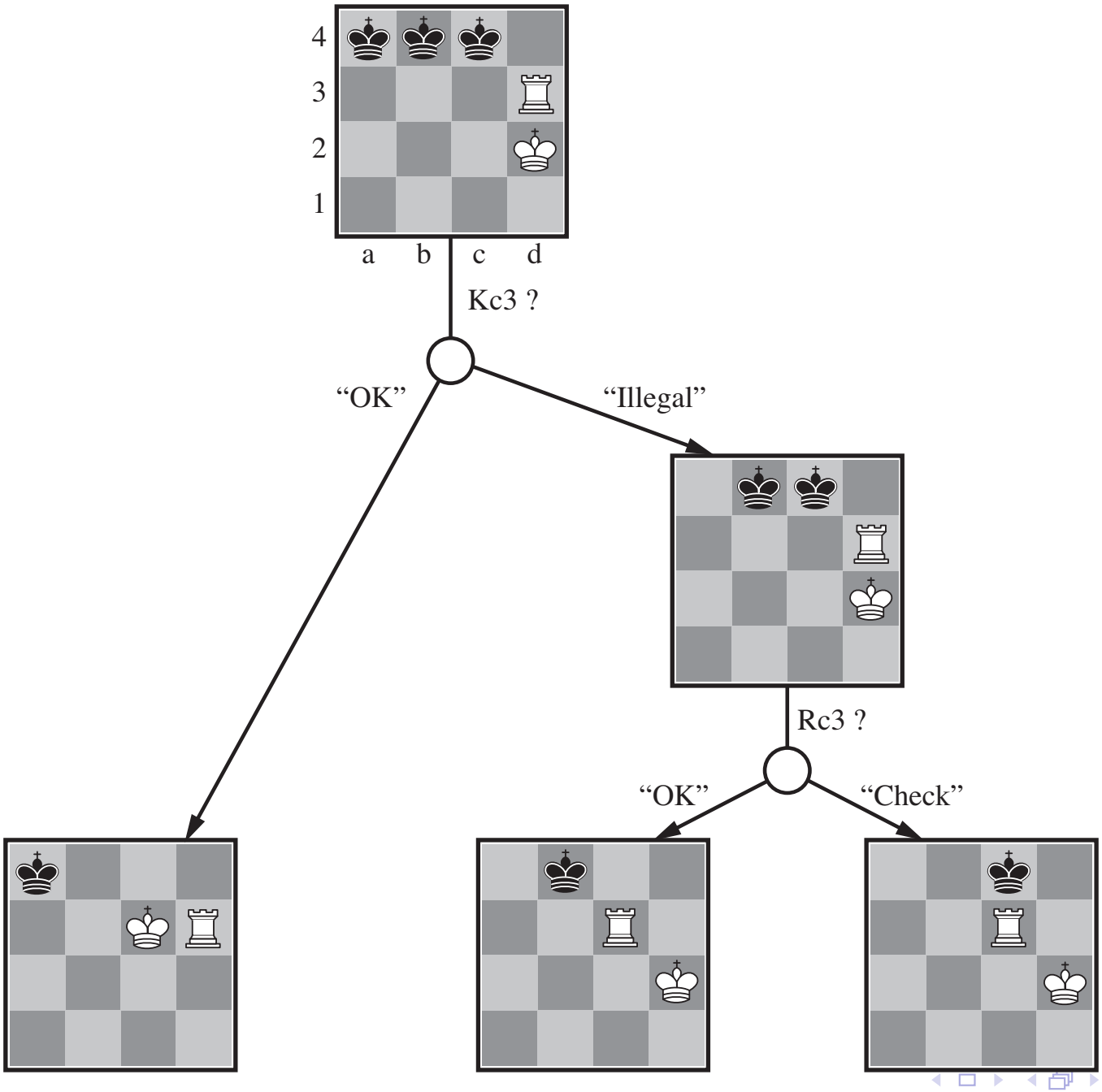
- Exemple: Battleship; jocuri de carti, unde cartile oponentului sunt necunoscute initial
- Se poate calcula probabilitatea fiecarei maini

IDEE: calculeaza valoare minimax a fiecarei actiuni in fiecare mana, dupa care alege actiunea cu cea mai mare valoare asteptata peste toate mainile

Sah fara a vedea mutarile adversarului- kriegspiel

The rules of Kriegspiel are as follows: White and Black each see a board containing only their own pieces. A referee, who can see all the pieces, adjudicates the game and periodically makes announcements that are heard by both players. First, White proposes to the referee a move that would be legal if there were no black pieces. If the black pieces prevent the move, the referee announces illegal, and White keeps proposing moves until a legal one is foundlearning more about the location of Blacks pieces in the process.
Once a legal move is proposed, the referee announces one or more of the following: Capture on square X if there is a capture, and Check by D if the black king is in check, where D is the direction of the check, and can be one of Knight, Rank, File, Long diagonal, or Short diagonal. If Black is checkmated or stalemated, the referee says so; otherwise, it is Blacks turn to move.

Sah fara a vedea mutarile adversarului- kriegspiel



Analiza

- Intuitia ca valoarea unei actiuni este media valorilor ei in toate starile este GRESITA
- In prezenta observablitatii partiale, valoarea unei actiuni depinde de starea de informare sau belief state a agentului
- Se poate genera si cauta un arbore a starilor de informatii

Conduce la comportament rational ca:

- realizarea de actiuni pentru a obtine informatii
- semnalarea unui partener
- actiuni random pentru a minimiza dezvaluirea de informatii

Rezumat

Strategii optimale

- Algoritmul MINIMAX
- $\alpha - \beta$ prunning

Decizii imperfecte, in timp real

Jocuri nedeterministe, cu sansa

Jocuri cu observabilitate partiala